The Discrete Fourier Transform

DFT equation, basis functions, spectra, real/complex sinusoids, and inverse DFT

Digital Signal Processing for Sound and Music · Topic 2

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DFT

Lecture roadmap

1 **DFT equation**
from samples to frequency bins

2 **DFT basis**
complex exponentials as analysis signals

3 **Interpretation**
magnitude, phase, bin frequency

4 **Sinusoids**
complex and real sinusoid spectra

5 **IDFT**
reconstructing time-domain signals

6 **Exercise 2**
implement sinusoids, DFT, IDFT, and magnitude spectrum

Goal: understand the DFT as a set of scalar products with complex sinusoids, and interpret the result as a spectrum.

The DFT: a change of viewpoint

Input: a finite sequence of samples

The signal is known only at N discrete time positions.

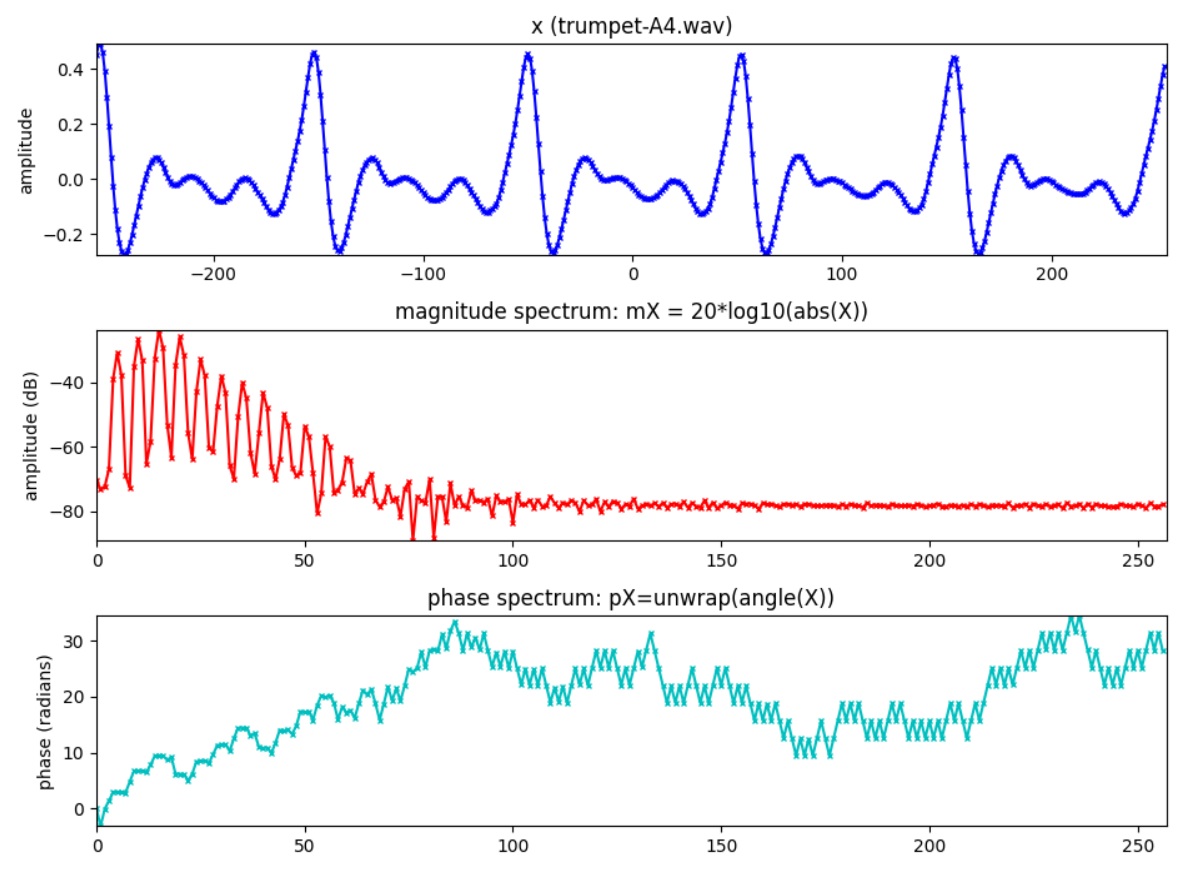
The DFT compares it against N analysis sinusoids.

The result is a complex number for each frequency bin.

Output: a spectrum

Magnitude tells how much of a bin frequency is present.

Phase tells how that sinusoidal component is aligned in time.



DFT equation and index meanings

$$X[k] = \sum_{n=0}^{N-1} x[n]e^{-j2\pi kn/N} \quad k = 0,\ldots,N-1$$

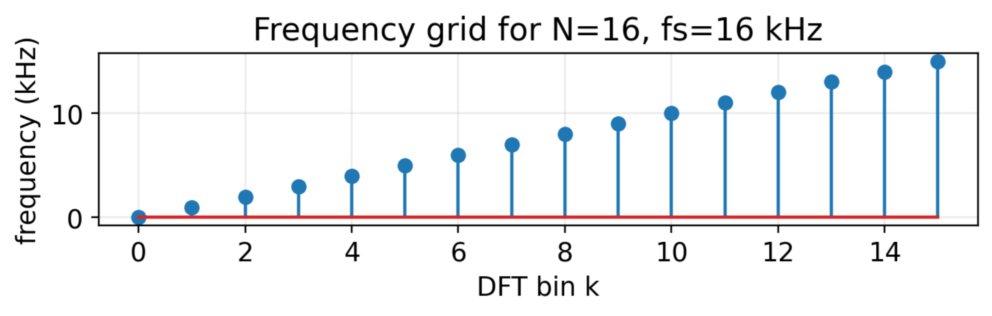
n: discrete time index(normalized time, $T = 1$)

k: discrete frequency index

$\omega_k = 2\pi\, k/N$: frequency in radians

$f_k = f_s\, k/N$: frequency in Hz(f_s: sampling rate)

Frequency index k maps each DFT bin to radians and to Hz.



Frequency bins are a fixed grid

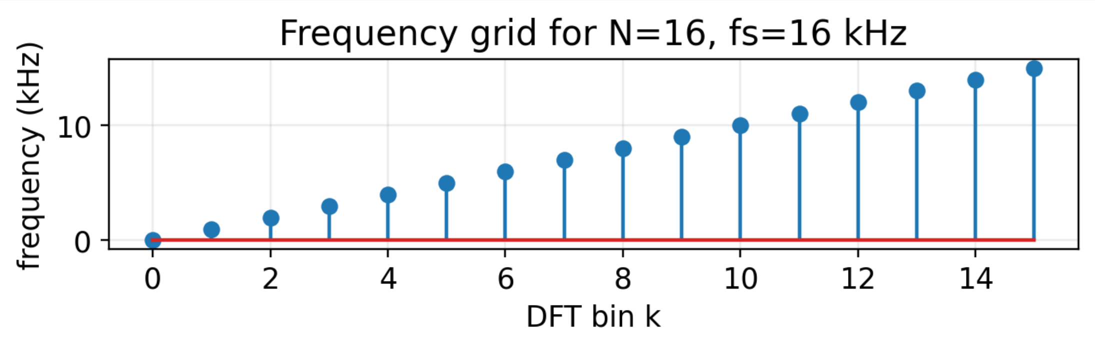
Bin frequency

$$f_k = \frac{k}{N} f_s, \quad \Delta f = \frac{f_s}{N}, \quad \omega_k = \frac{2\pi k}{N}$$

The DFT does not test all possible frequencies. It tests exactly N frequencies determined by N and f_s.

Increasing N gives a finer frequency grid.

A sinusoid exactly on a bin gives a sharp spectral peak; off-bin frequencies spread across bins.



DFT basis functions: complex exponentials

$$s_k^* = e^{-j2\pi kn/N} = cos(2\pi\, kn/N) - jsin(2\pi\, kn/N)$$

for $N = 4$, thus for $n = 0,1,2,3;\ k = 0,1,2,3$

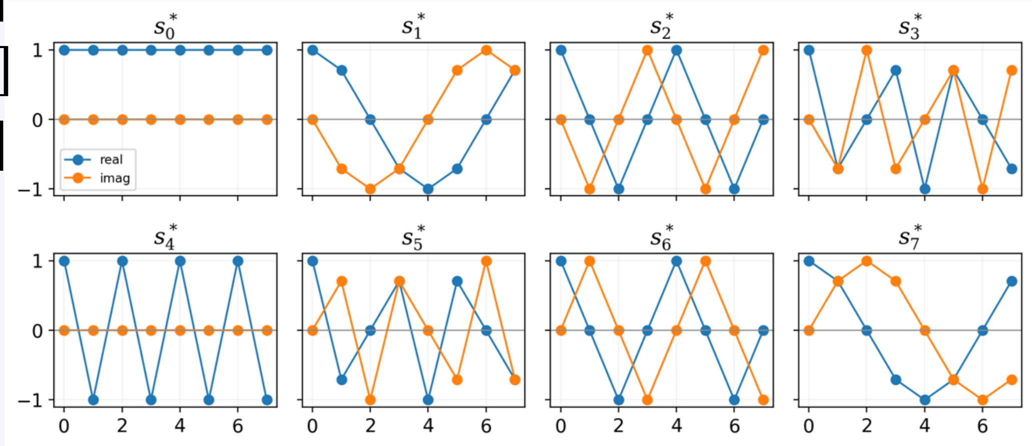
$$s_0^* = cos(2\,\pi \times 0 \times n/4) - jsin(2\,\pi \times 0 \times n/4) = [1,1,1,1]$$
$$s_1^* = cos(2\,\pi \times 1 \times n/4) - jsin(2\,\pi \times 1 \times n/4) = [1,-j,-1,j]$$
$$s_2^* = cos(2\,\pi \times 2 \times n/4) - jsin(2\,\pi \times 2 \times n/4) = [1,-1,1,-1]$$
$$s_3^* = cos(2\,\pi \times 3 \times n/4) - jsin(2\,\pi \times 3 \times n/4) = [1,j,-1,-j]$$

Why these signals?

For each k, the DFT uses one complex exponential.

The conjugate basis s*k rotates in the opposite direction.

For N = 4 the four basis sequences can be written explicitly.



Plots for N = 8.

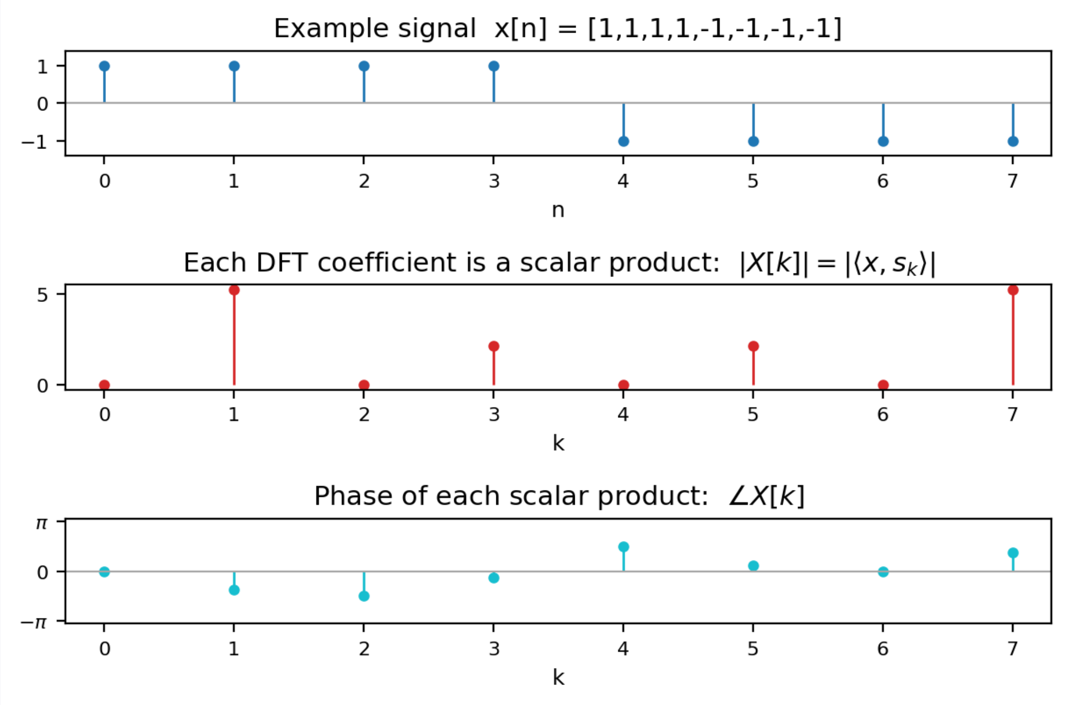
The DFT as scalar products

$$\langle x, s_k \rangle = \sum_{n=0}^{N-1} x[n] s_k^*[n] = \sum_{n=0}^{N-1} x[n] e^{-j2\pi kn/N}$$

What the formula means

Top: example signal x[n].

Middle: magnitude abs(X[k]).
Bottom: phase angle(X[k]).
Each k corresponds to one scalar product.



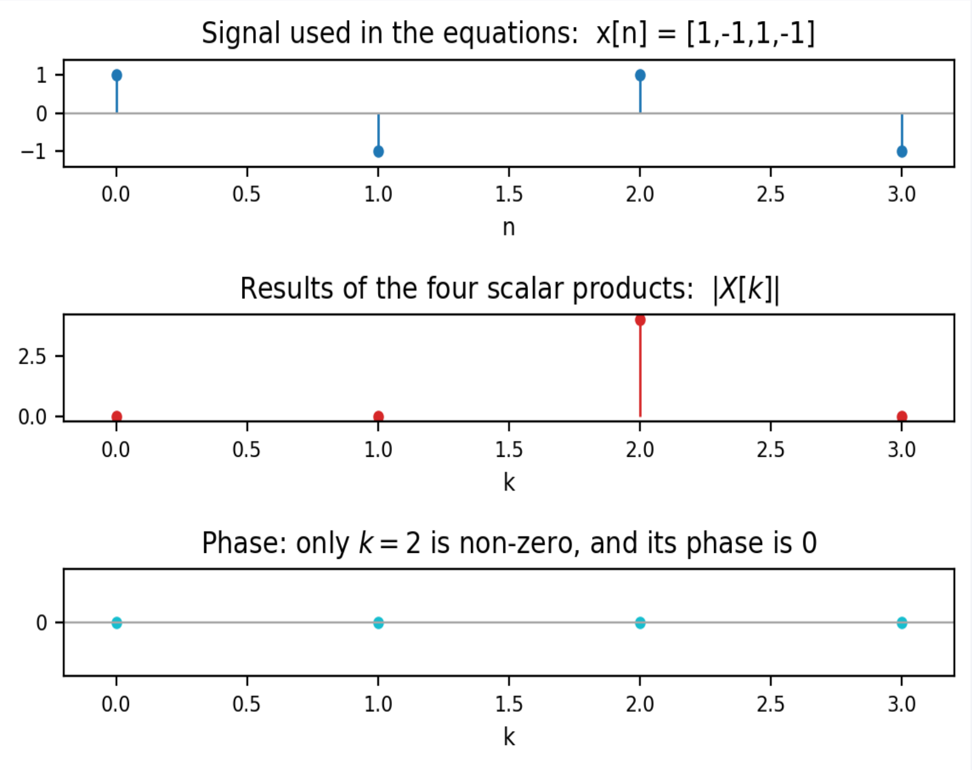
Example: N = 4 scalar products

Worked example directly matched to the equations

$$x[n] = [1, -1, 1, -1]; N = 4$$

$$\langle x, s_0 \rangle = 1 \times 1 + (-1) \times 1 + 1 \times 1 + (-1) \times 1 = 0$$
$$\langle x, s_1 \rangle = 1 \times 1 + (-1) \times (-j) + 1 \times (-1) + (-1) \times j = 0$$
$$\langle x, s_2 \rangle = 1 \times 1 + (-1) \times (-1) + 1 \times 1 + (-1) \times (-1) = 4$$
$$\langle x, s_3 \rangle = 1 \times 1 + (-1) \times j + 1 \times (-1) + (-1) \times (-j) = 0$$

The plot shows the same result as the equations: X[0]=0, X[1]=0, X[2]=4, X[3]=0.



Magnitude and phase: what does the spectrum store?

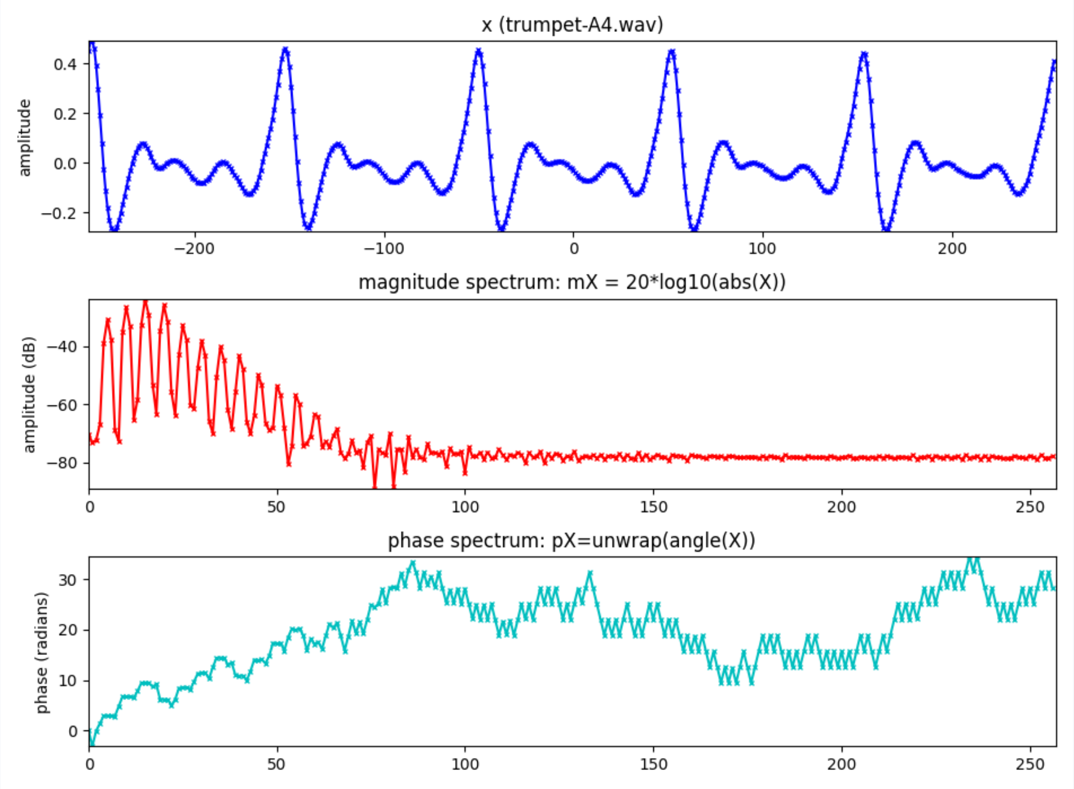
A DFT value is complex:

$$X[k] = |X[k]|e^{j\angle X[k]}$$

Magnitude |X[k]|: strength of the kth frequency component.

Phase ∠X[k]: temporal alignment of that component.

For audio, magnitude is often easier to interpret; phase is essential for reconstruction.



DFT of a complex sinusoid exactly on a bin

$x_1[n] = e^{j2\pi k_0 n/N}$ for $n = 0, \ldots, N-1$

$$X_1[k] = \sum_{n=0}^{N-1} x_1[n] e^{-j2\pi kn/N}$$

$$= \sum_{n=0}^{N-1} e^{j2\pi k_0 n/N}\, e^{-j2\pi kn/N}$$

$$= \sum_{n=0}^{N-1} e^{-j2\pi (k-k_0)n/N}$$

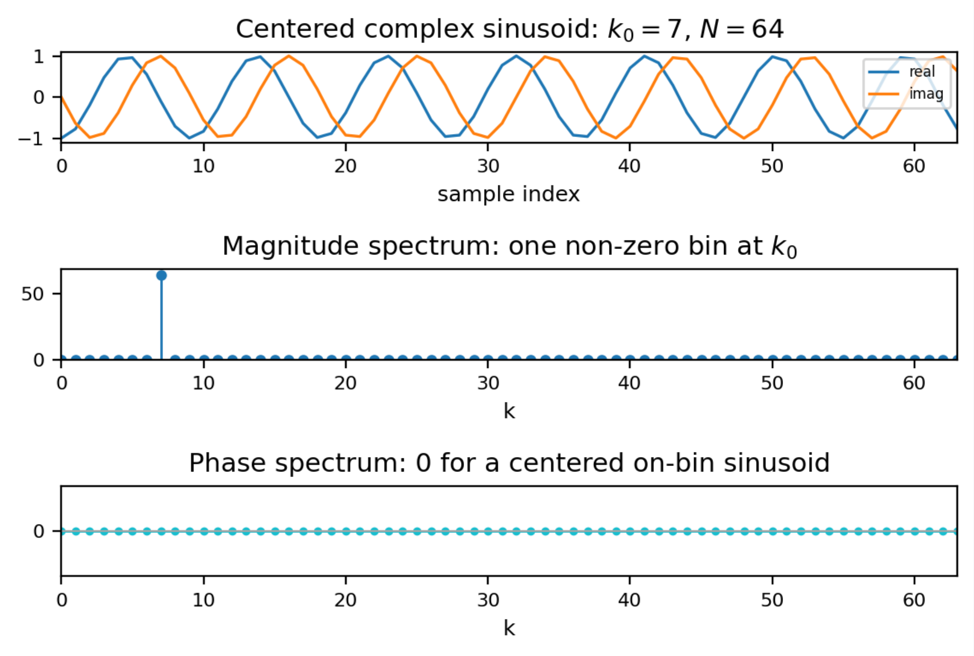
$$= \frac{1 - e^{-j2\pi(k-k_0)}}{1 - e^{-j2\pi(k-k_0)/N}} \text{ (sum of a geometric series)}$$

if $k \neq k_0$, denominator $\neq 0$ and numerator $= 0$

thus $X_1[k] = N$ for $k = k_0$ and $X_1[k] = 0$ for $k \neq k_0$

Main result

All energy is at bin k_0.
For the centered signal, the phase is 0.



DFT of any complex sinusoid

$$x_2[n] = e^{j(2\pi f_0 n+\phi)} \text{ for } n = 0, \ldots, N-1$$

$$X_2[k] = \sum_{n=0}^{N-1} x_2[n] e^{-j2\pi kn/N}$$

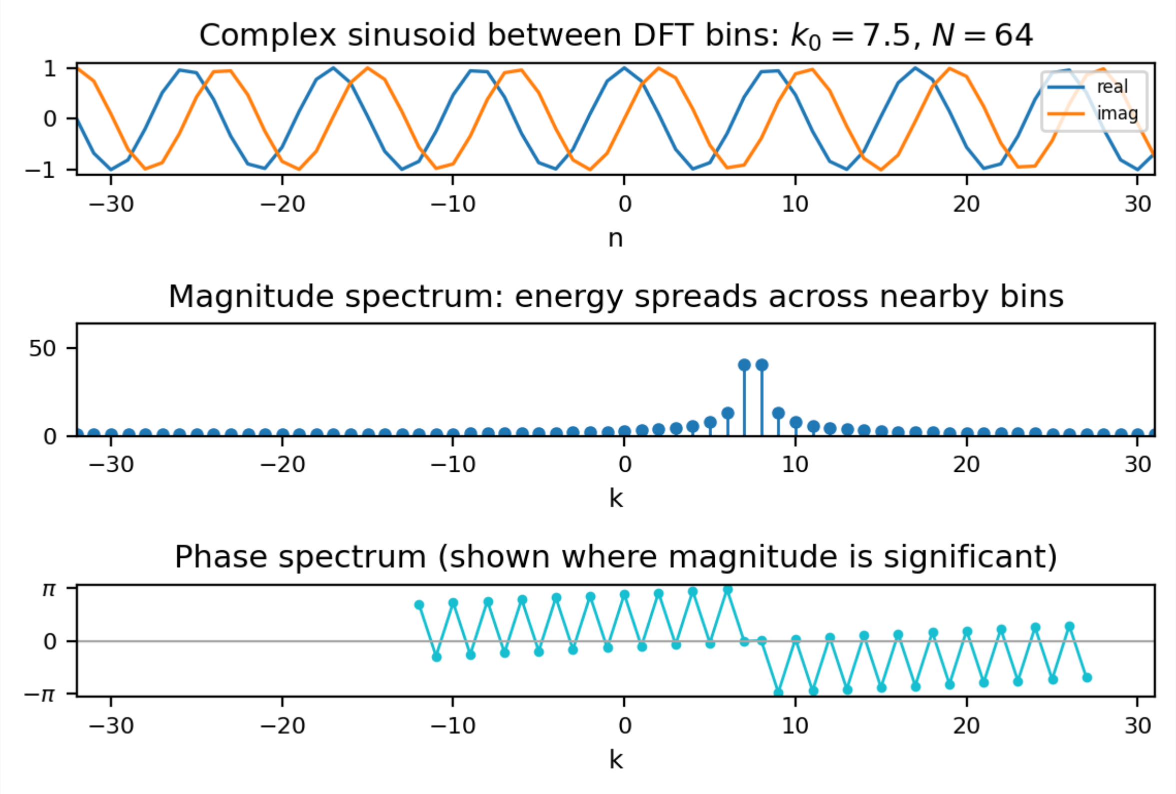
$$= \sum_{n=0}^{N-1} e^{j(2\pi f_0 n+\phi)}\, e^{-j2\pi kn/N}$$

$$= e^{j\phi} \sum_{n=0}^{N-1} e^{-j2\pi(k/N-f_0)n}$$

$$= e^{j\phi} \frac{1-e^{-j2\pi(k/N-f_0)N}}{1-e^{-j2\pi(k/N-f_0)}}$$

Off-bin case

Off-bin sinusoids spread energy across bins.
The phase spectrum is now also shown.



DFT of real sinusoids

$$x_3[n] = A_0 cos(2\pi k_0\, n/N) = \frac{A_0}{2} e^{j2\pi k_0 n/N} + \frac{A_0}{2} e^{-j2\pi k_0 n/N}$$

$$X_3[k] = \sum_{n=-N/2}^{N/2-1} x_3[n] e^{-j2\pi kn/N}$$

$$= \sum_{n=-N/2}^{N/2-1} \left(\frac{A_0}{2} e^{j2\pi k_0 n/N} + \frac{A_0}{2} e^{-j2\pi k_0 n/N}\right) e^{-j2\pi kn/N}$$

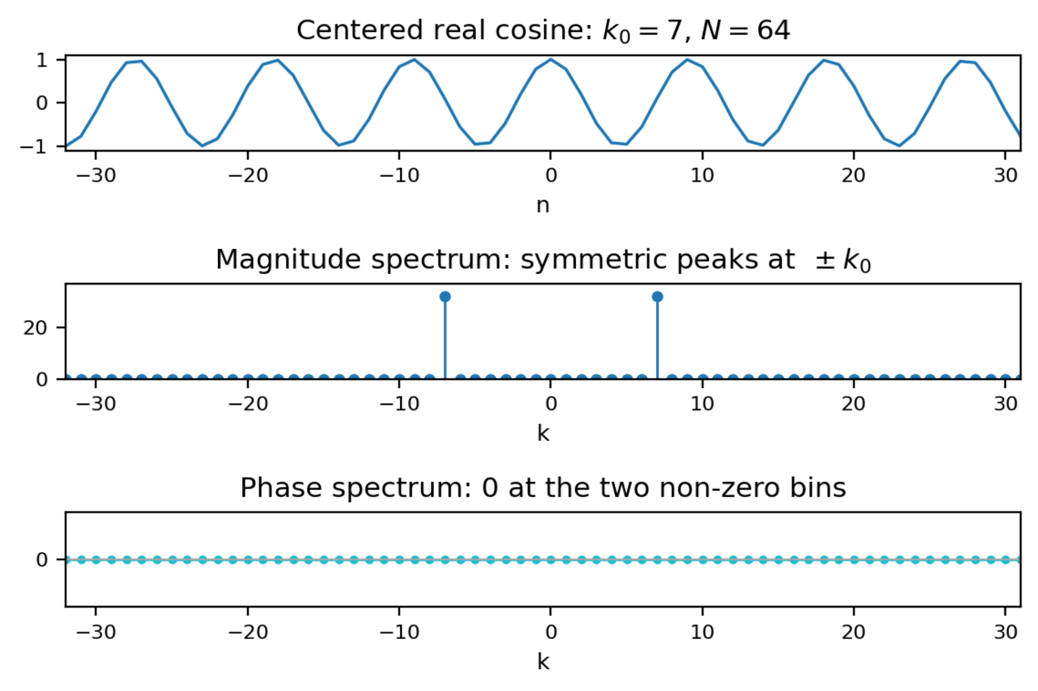
$$= \sum_{n=-N/2}^{N/2-1} \frac{A_0}{2} e^{j2\pi k_0 n/N} e^{-j2\pi kn/N} + \sum_{n=-N/2}^{N/2-1} \frac{A_0}{2} e^{-j2\pi k_0 n/N} e^{-j2\pi kn/N}$$

$$= \sum_{n=-N/2}^{N/2-1} \frac{A_0}{2} e^{-j2\pi (k-k_0) n/N} + \sum_{n=-N/2}^{N/2-1} \frac{A_0}{2} e^{-j2\pi (k+k_0) n/N}$$

$$= N\frac{A_0}{2} \text{for} k = k_0, -k_0; 0 \text{ for rest of } k$$

Positive + negative frequencies

A centered real cosine has two symmetric peaks. Their phases are 0.



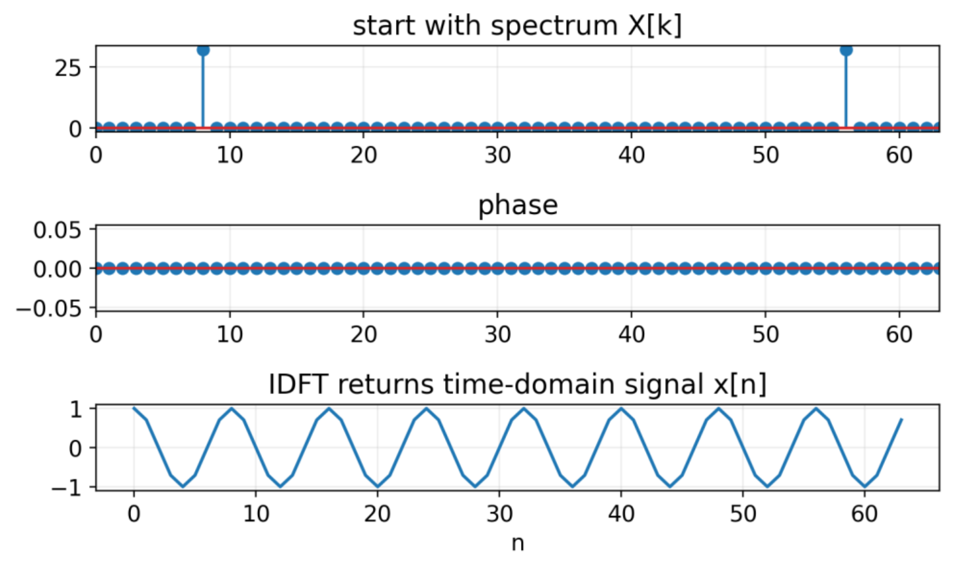
Inverse DFT: from spectrum back to samples

$$x[n] = \frac{1}{N}\sum_{k=0}^{N-1} X\,[k] s_k[n] = \frac{1}{N}\sum_{k=0}^{N-1} X\,[k] e^{j2\pi kn/N}$$
$$n \quad = 0,1,\ldots,N-1$$

The IDFT reconstructs x[n] by summing all basis functions weighted by X[k].

$$X[k] = [0,4,0,0]; N = 4$$

$$x[0] = \frac{1}{4}(X * s)[n=0] = \frac{1}{4}(0*1+4*1+0*1+0*1) = 1$$
$$x[1] = \frac{1}{4}(X * s)[n=1] = \frac{1}{4}\big(0*1+4*j+0*(-1)+0*(-j)\big) = j$$
$$x[2] = \frac{1}{4}(X * s)[n=2] = \frac{1}{4}\big(0*1+4*(-1)+0*1+0*(-1)\big) = -1$$
$$x[3] = \frac{1}{4}(X * s)[n=3] = \frac{1}{4}(0*1+4*(-j)+0*(-1)+0*j) = -j$$

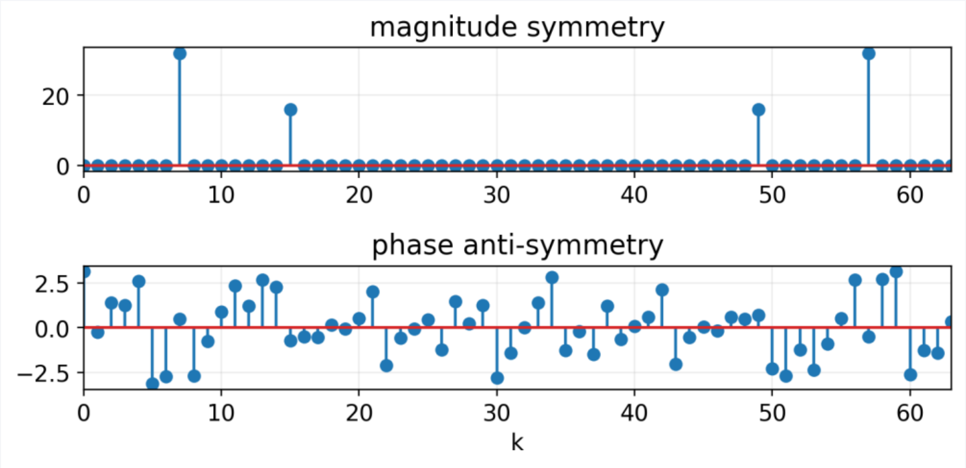


Inverse DFT for real signals

$$X[k] = |X[k]|e^{j<X[k]} \text{ and } X[-k] = |X[k]|e^{-j<X[k]}$$
$$\text{for } k = 0,1,\ldots,N/2$$

Conjugate symmetry

For real-valued time signals, positive and negative frequency coefficients are complex conjugates: equal magnitudes and opposite phases.

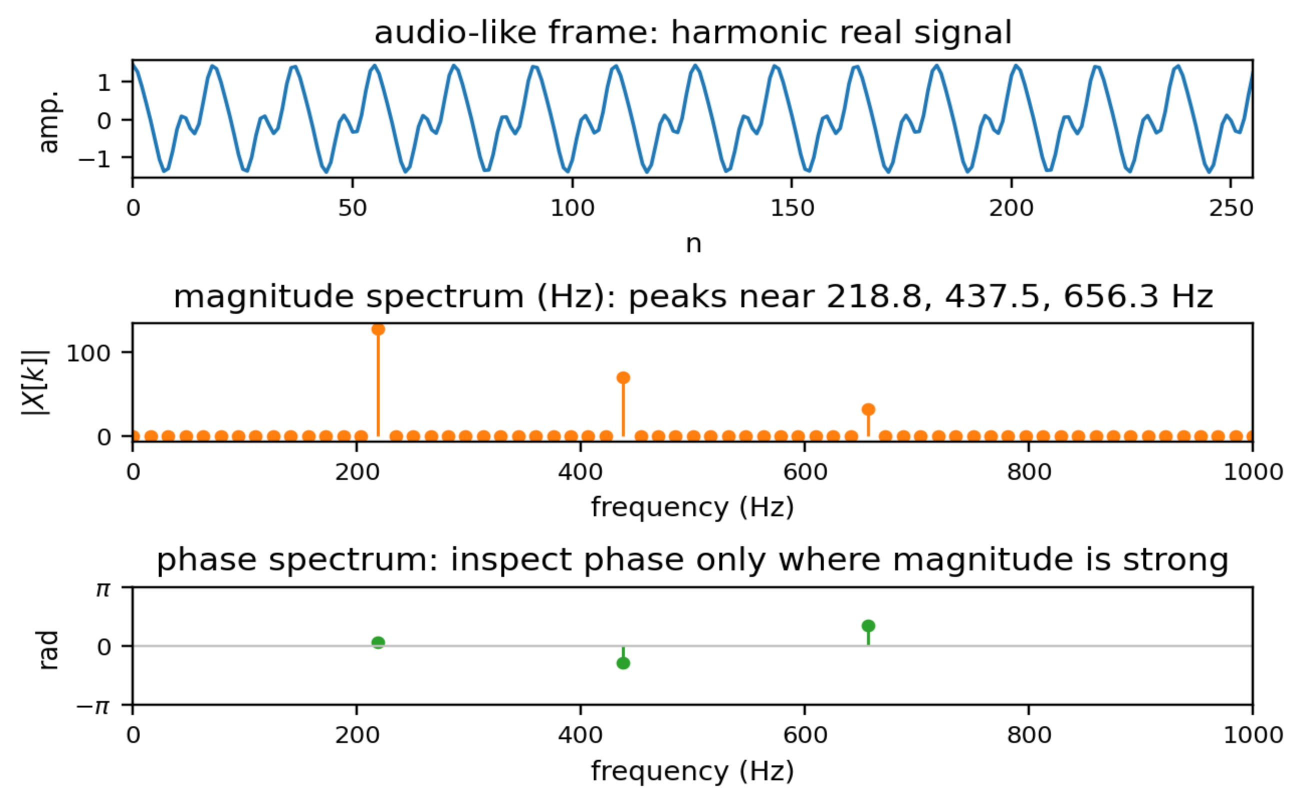


Practical interpretation for audio signals

When reading a spectrum, ask:

- What is the sampling rate f_s?
- What is the DFT size N?
- Which bin k corresponds to the frequency I care about?
- Is the sinusoid exactly at a bin center?
- Am I looking at magnitude, phase, or both?

These questions determine whether a peak is easy to interpret, smeared across bins, or mirrored by real-signal symmetry.



What to remember from Topic 2

1. **DFT equation**
   computes one complex coefficient per frequency bin

2. **Basis functions**
   complex exponentials indexed by k

3. **Scalar products**
   measure similarity between x[n] and each basis function

4. **Complex sinusoids**
   one bin if on-grid, leakage if off-grid

5. **Real sinusoids**
   two symmetric peaks

6. **IDFT**
   reconstructs samples by summing all components

EXAMPLE

Example notebook: DFT analysis / resynthesis workflow

examples/dft-examples.ipynb turns the DFT equation into a concrete pipeline: read a sound, analyze one frame, and synthesize it back.

1 Load sound

read audio and listen

2 Select frame

choose a short segment

3 Window + DFT

compute magnitude + phase

4 IDFT synth

reconstruct the frame

Key code block

```
window = 'hamming'
M = 256      # analysis-window size
N = 1024     # DFT size
frame_time = 0.2

w = get_window(window, M)
x_frame = x[start_sample:start_sample + M]
mX, pX = DFT.dftAnal(x_frame, w, N)
y_frame = DFT.dftSynth(mX, pX, w.size)
```

What students should notice

- M controls how much time context is analyzed.
- N controls the spacing between DFT frequency bins.
- mX and pX are the magnitude and phase spectra.
- Both magnitude and phase are needed for resynthesis.

EXAMPLE

What the DFT example shows

The notebook links abstract DFT coefficients to visible and audible results: a selected frame, its spectrum, and a resynthesized waveform.

1 · Analysis

Isolate a short frame, multiply it by a window, and transform it into magnitude and phase.

2 · Interpretation

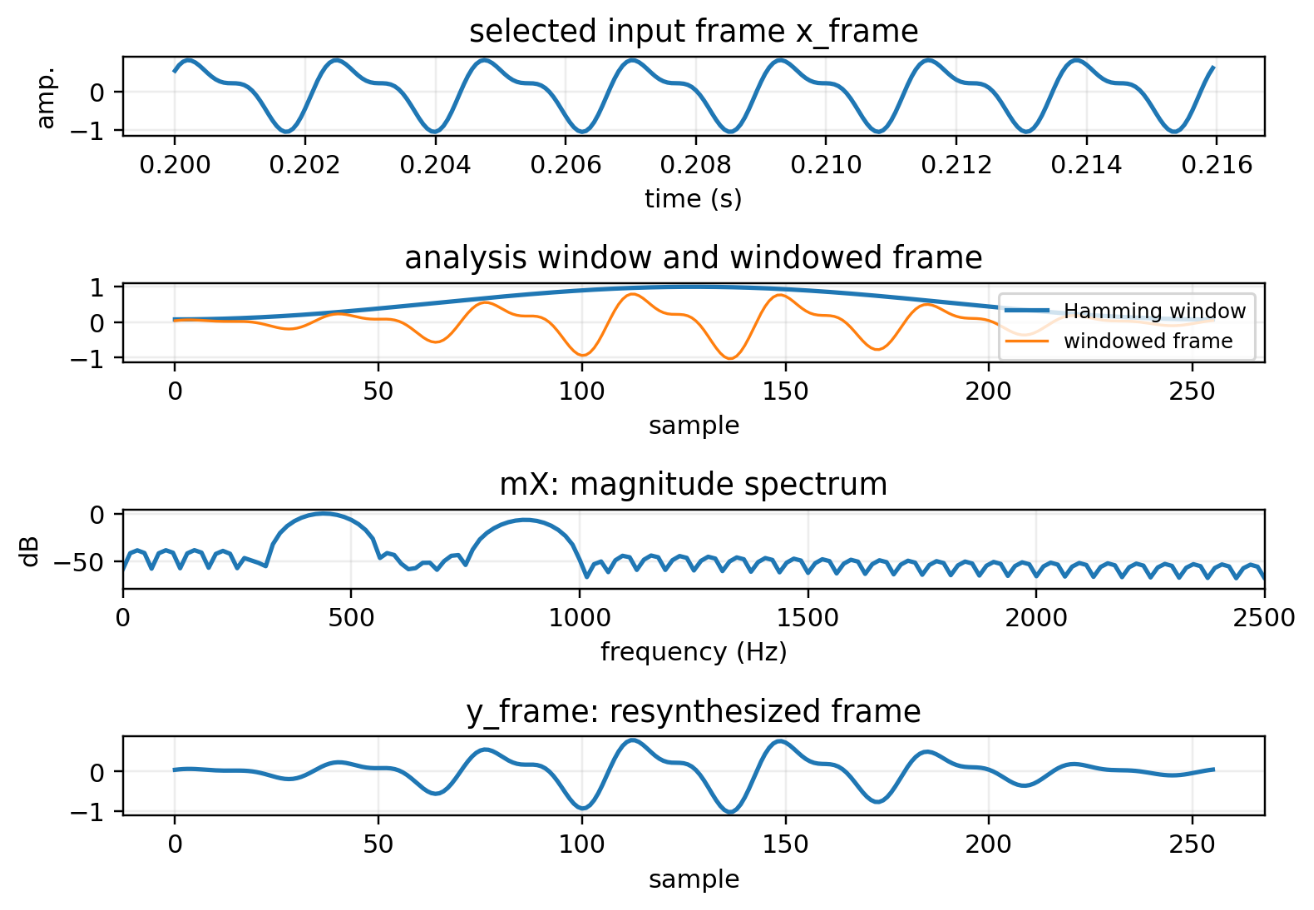
Peaks in mX indicate strong sinusoidal components; pX gives their phase alignment.

3 · Resynthesis

Use the DFT coefficients to reconstruct the time-domain frame.

4 · Parameters

M fixes the frame duration; N fixes the DFT frequency grid used to analyze it.



LAB

Exercise 2: Sinusoids and the DFT

Second lab: turn DFT formulas into code, plots, spectra, and reconstruction.

What you will practice

- generate and inspect real sinusoids
- build DFT complex sinusoids for bin k
- implement DFT, IDFT, and magnitude spectra
- connect spectrum peaks with the signal you created

How to start

Open `exercises/E2-Sinusoids-and-DFT.ipynb`

Use: NumPy · Matplotlib · complex exponentials · plots

Compare with: examples/dft-examples.ipynb

Lab structure

1 · Real sinusoid
- A, f, ϕ, fs, duration
- time plot + listening

2 · DFT basis
- complex sinusoid
- bin k, length N

3 · DFT
- scalar products
- one X[k] per bin

4 · IDFT
- sum weighted bases
- include 1/N scaling

5 · Spectrum
- magnitude peaks
- interpret bin locations

Main check: can you predict the DFT peaks before computing them, and can the IDFT reconstruct the samples?

References and credits

- MTG sms-tools-materials repository: lecture slides, examples, exercises, and sounds — https://github.com/MTG/sms-tools-materials
- Exercise 2 notebook: E2-Sinusoids-and-DFT.ipynb
- Julius O. Smith, Mathematics of the Discrete Fourier Transform — https://ccrma.stanford.edu/~jos/mdft/
- Discrete Fourier Transform overview — https://en.wikipedia.org/wiki/Discrete_Fourier_transform
- Sounds from Freesound and sms-tools-materials sounds folder.